Week 7 Reading

Sensitivity & Value of Information

Draft Material — This content is under development and subject to change.

1 Assigned Readings

1. A. H. Murphy, R. W. Katz, R. L. Winkler, and W.-R. Hsu [1] – "Repetitive Decision Making and the Value of Forecasts in the Cost-Loss Ratio Situation." Introduces the cost-loss framework for quantifying the economic value of weather forecasts, extending from a static to a dynamic (multi-period) setting.
2. M. C. Runge, S. J. Converse, and J. E. Lyons [2] – "Which Uncertainty?" Uses expected value of perfect information (EVPI) and partial EVPI to identify which uncertainties most impede management of endangered whooping cranes. A clear worked example of VOI applied to real decision-making.

2 Discussion Questions

1. In the Runge et al. case study, "restore meadows" is the best strategy under uncertainty, but it is not the best strategy under *any* individual hypothesis. How is this possible? What does this imply about decision-making under uncertainty?
2. The partial EVPI analysis shows that resolving the "black fly" hypothesis captures 54% of the total information value. What does this mean for how the wildlife refuge should allocate its monitoring budget?
3. In Murphy et al.'s dynamic cost-loss model, the decision threshold changes from C/L (static) to $C/(L-C)$ (two-period). Why does anticipating the future raise the threshold for protection? What's the intuition?
4. Murphy et al. show that the value of imperfect forecasts is always non-negative (Blackwell's theorem). Can you think of a real-world situation where acting on a forecast made things worse? How do you reconcile this with the theorem?

Bibliography

[1] A. H. Murphy, R. W. Katz, R. L. Winkler, and W.-R. Hsu, "Repetitive Decision Making and the Value of Forecasts in the Cost-Loss Ratio Situation: A Dynamic Model," *Monthly Weather Review*, vol. 113, no. 5, pp. 801–813, May 1985, doi: 10.1175/1520-0493(1985)113<0801:RDMATV>2.0.CO;2.

[2] M. C. Runge, S. J. Converse, and J. E. Lyons, "Which Uncertainty? Using Expert Elicitation and Expected Value of Information to Design an Adaptive Program," *Biological Conservation*, vol. 144, no. 4, pp. 1214–1223, Apr. 2011, doi: 10.1016/j.biocon.2010.12.020.